Lecture 28.04: Gini Impurity

📌 Topic Overview

In this lecture, we study the second major impurity metric: **Gini Impurity**. This is the default metric used by CART and Scikit-Learn. We examine who invented it, its formula, range, and blackboard calculation.

🏛️ What is Gini Impurity?

Named after the Italian statistician and demographer **Corrado Gini**, Gini Impurity measures:

> **The likelihood of a random element being misclassified if it were randomly labeled according to the class distribution in the dataset.**

The general formula is:

$$GI = 1 - \sum_{i=1}^{c} (p_i)^2$$

For binary classification (`Yes` and `No`):

$$GI = 1 - (p_{extyes}^2 + p_{extno}^2)$$

📊 Range of Gini Impurity for Binary Classification

- **Minimum Value = 0.0**: When all samples belong to a single class (completely pure).
- **Maximum Value = 0.5**: When classes are equally split (50% Yes and 50% No).

$$extMaxGI = 1 - (0.5^2 + 0.5^2) = 1 - (0.25 + 0.25) = 1 - 0.5 = 0.5$$

> **Key Distinction**: While maximum Entropy is **1.0**, maximum Gini Impurity for binary classification is **0.5**!

🧮 Numerical Examples from Blackboard

Example 1: Impure Node $[3n, 3y]$

- Total samples = 6
- p_y = ↑rac{3}{6} = 0.5, p_n = ↑rac{3}{6} = 0.5

> GI = 1 - (p_y^2 + p_n^2) = 1 - \left(↑rac{1}{4} + ↑rac{1}{4} ight) = 1 - ↑rac{1}{2} = 0.5

Example 2: Pure Node $[0n, 3y]$

- Total samples = 3
- p_y = ↑rac{3}{3} = 1.0, $p_n = 0.0$

> $$GI = 1 - (1.0^2 + 0.0^2) = 1 - 1 = 0.0$$

📈 Comparing Curves: Entropy vs Gini Impurity

On the blackboard, the instructor plots both curves on the same axes:

- The **orange curve (Entropy)** peaks at **1.0** when $p = 0.5$.
- The **blue curve (Gini Impurity)** peaks at **0.5** when $p = 0.5$.
- Both curves share the exact same minima (0.0) at $p = 0$ and $p = 1$.

Because both curves have virtually identical shapes, they almost always agree on which feature split is best!

💻 Python Implementation

```
def calculate_gini_impurity(yes_count, no_count):
    """
```

```
    Calculates Gini Impurity for a binary classification node.
    Formula: GI(S) = 1 - (p_yes^2 + p_no^2)

    Returns:
        float: Gini value in range [0.0 (Pure), 0.5 (Maximum Impurity)]
    """
    total = yes_count + no_count
    if total == 0:
        return 0.0

    p_yes = yes_count / total
    p_no = no_count / total

    gini = 1.0 - (p_yes**2 + p_no**2)
    return gini

# --- Function Call & Verification ---
print("Gini [3 Yes, 3 No]:", calculate_gini_impurity(3, 3))  # 0.5 (Maximum Impuri
print("Gini [0 Yes, 3 No]:", calculate_gini_impurity(0, 3))  # 0.0 (Pure node)
print("Gini [2 Yes, 4 No]:", round(calculate_gini_impurity(2, 4), 3))  # 0.444
```

🔑 Key Takeaways

1. Gini Impurity measures the probability of misclassification.
2. For binary classification, Gini ranges from **0.0 (pure)** to **0.5 (maximum impurity)**.
3. Because Gini only requires squaring numbers (no logarithms), it is computationally faster than Entropy.